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Annexe 2
Appendix 2

Valeurs recommandées des fréquences étalons destinées à la mise en pratique de la définition du mètre et aux représentations secondaires de la seconde

Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second

Partie 2.20
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Strontium 87 Atom ($f \approx 429\,\mathrm{THz}$)

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August 2018
août 2018

Bureau International
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Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second Part 2.20: Strontium 87 Atom ($f \approx 429\,\mathrm{THz}$)

9th edition 2019

^{87}Sr neutral atom, $5s^2\,{}^1S_0 - 5s5p\,{}^3P_0$ unperturbed optical transition

1. Recommended value [1] of the frequency in the CIPM List of Frequencies

$f(^{87}\mathrm{Sr}) = 429\,228\,004\,229\,873.0\ \mathrm{Hz}$

equivalent to

$\lambda(^{87}\mathrm{Sr}) = 698\,445\,709.612\,754\,4\ \mathrm{fm},$

with a relative standard uncertainty of 4×10^{-16}.

This radiation was endorsed by the CIPM as a secondary representation of the definition of the second [2].

2. Source data

The adopted value of $f\left(^{87}\mathrm{Sr}\right) = 429\,228\,004\,229\,873.0$ Hz with a relative standard uncertainty of 4×10^{-16}

has been calculated by the methods presented in [3], [4], [5], [6], with the data from the table of frequencies (2017) attached below. The recommended uncertainty is the same as that of the most accurate comparison between the Cs Primary Frequency Standards at LNE-SYRTE and PTB via an optical fibre link [7] i.e. 4×10^{-16}.

Table 1. Table of evaluated frequencies (2017)

Frequency / Hz	Uncertainty / Hz	Reference
^{199}Hg		
1 128 575 290 808 155.1	6.4	[7]
1 128 575 290 808 154.62	0.41	[8]
^{171}Yb		
518 295 836 590 864.0	28.0	[38]
518 295 836 590 863.1	2.0	[39]
518 295 836 590 865.2	0.7	[40]
518 295 836 590 863.5	8.1	[41]
518 295 836 590 863.59	0.31	[42]
518 295 836 590 863.38	0.57	[43]
^{88}Sr		
429 228 066 418 009.0	32.0	[9]
429 228 066 418 008.3	2.1	[10], Sr1
429 228 066 418 007.3	2.9	[10], Sr2
^{87}Sr		
429 228 004 229 874.0	1.1	[11]
429 228 004 229 873.65	0.37	[12]
429 228 004 229 873.6	1.1	[13]
429 228 004 229 874.1	2.4	[14]
429 228 004 229 872.9	0.5	[15]
429 228 004 229 873.9	1.4	[16]
429 228 004 229 872.0	1.6	[17]
429 228 004 229 873.56	0.49	[18]
429 228 004 229 873.7	1.4	[19]
429 228 004 229 873.13	0.17	[20]
429 228 004 229 873.10	0.13	[21]
429 228 004 229 872.92	0.12	[22]
429 228 004 229 872.97	0.16	[23], 2014
429 228 004 229 873.04	0.11	[23], June 2015
429 228 004 229 872.97	0.40	[24]
429 228 004 229 872.99	0.18	[25]

Frequency / Hz	Uncertainty / Hz	Reference
$^{40}Ca^{+}$		
411 042 129 776 393.2	3.0 [(a)]	[26]
411 042 129 776 398.4	1.2	[27]
411 042 129 776 400.5	1.2	[28], 2012
411 042 129 776 401.7	1.1	[28], 2014/2015
^{87}Rb		
6 834 682 610.904 312 9	2.4×10^{-6}	[29]
6 834 682 610.904 307 0	3.1×10^{-6}	[30]
6 834 682 610.904 312 5	2.1×10^{-6}	[31]
Frequency ratios	Fractional uncertainty	Reference
$^{199}Hg/^{87}Sr$		
2.629 314 209 898 909 60	8.4×10^{-17}	[32]
2.629 314 209 898 909 15	1.8×10^{-16}	[8]
$^{199}Hg/^{87}Rb$		
165 124.754 879 997 258	3.8×10^{-16}	[8]
$^{171}Yb/^{87}Sr$		
1.207 507 039 343 341 2	1.4×10^{-15}	[33], [34]
1.207 507 039 343 337 76	2.4×10^{-16}	[35]
1.207 507 039 343 337 749	4.6×10^{-17}	[36]
$^{88}Sr/^{87}Sr$		
1.000 000 144 883 682 777	7×10^{-17} [(b)]	[37]
$^{87}Sr/^{40}Ca^{+}$		
1.044 243 334 529 641 6	2.3×10^{-15}	[27]
$^{87}Sr/^{87}Rb$		
62 801.453 800 512 435	3.3×10^{-16}	[22]

(a) From the least square procedure it turned out that the value and the uncertainty given in ref. [26] is not compatible with the remaining data. Thus the uncertainty given in the original publication [26] was increased to 3 Hz to make it statistically more consistent.

(b) The fractional uncertainty of 2.3×10^{-17} in ref. [37] has been increased by a factor of three since this uncertainty is an order of magnitude smaller than the other input data for this transition.

References

[1] Consultative Committee for Time and Frequency (CCTF): Report of the 21st meeting (8-9 June 2017) to the International Committee for Weights and Measures; https://www.bipm.org/utils/common/pdf/CC/CCTF/CCTF21.pdf

[2] Recommendation 1 (CI-2006) https://www.bipm.org/utils/en/pdf/CIPM/CIPM2006-EN.pdf

[3] H. S. Margolis, P. Gill: Least-squares analysis of clock frequency comparison data to deduce optimized frequency and frequency ratio values; *Metrologia* **52**, 628 (2015)

[4] L. Robertsson: On the evaluation of ultra-high-precision frequency ratio measurements: examining closed loops in a graph theory framework; *Metrologia* **53**, 1272 (2016)

[5] Chris Oates: private communication. An independent program was developed in Mathematica at NIST (2017)

[6] F. Riehle, P. Gill, F. Arias, L. Robertsson: The CIPM List of Recommended Frequency Standard Values: Guidelines and Procedures; *Metrologia* **55**, 188-200 (2018)

[7] J. J. McFerran, L. Yi, S. Mejri, S. Di Manno, W. Zhang, J. Guéna, Y. Le Coq, S. Bize: Erratum: Neutral Atom Frequency Reference in the Deep Ultraviolet with Fractional Uncertainty $= 5.7 \times 10^{-15}$ [*Phys. Rev. Lett.* **108**, 183004 (2012)]; *Phys. Rev. Lett.* **115**, 219901 (2015)

[8] R. Tyumenev, M. Favier, S. Bilicki, E. Bookjans, R. Le Targat, J. Lodewyck, D. Nicolodi, Y. Le Coq, M. Abgrall, J. Guéna, L. De Sarlo, S. Bize: Comparing a mercury optical lattice clock with microwave and optical frequency standards; *New J. Phys.* **18**, 113002 (2016)

[9] X. Baillard, M. Fouché, R. Le Targat, P. G. Westergaard, A. Lecallier, Y. Le Coq, G. D. Rovera, S. Bize, P. Lemonde: Accuracy evaluation of an optical lattice clock with bosonic atoms; *Opt. Lett.* **32** 1812 (2007).

[10] P. Morzyński, M. Bober, D. Bartoszek-Bober, J. Nawrocki, P. Krehlik, Ł. Śliwczyński, M. Lipiński, P. Masłowski, A. Cygan, P. Dunst, M. Garus, D. Lisak, J. Zachorowski, W. Gawlik, C. Radzewicz, R. Ciuryło, M. Zawada: Absolute measurement of the $^1S_0 - {}^3P_0$ clock transition in neutral ^{88}Sr over the 330 km-long stabilized fibre optic link; *Scientific Reports* **5**, 17495 (2015)

[11] M. M. Boyd, A. D. Ludlow, S. Blatt, S. M. Foreman, T. Ido, T. Zelevinsky, J. Ye.: ^{87}Sr lattice clock with inaccuracy below 10-15; *Phys. Rev. Lett.* **98**, 083002 (2007)

[12] G. K. Campbell, A. D. Ludlow, S. Blatt, J. W. Thomsen, M. J. Martin, M. H. G. de Miranda, T. Zelevinsky, M. M. Boyd, J. Ye, S. A. Diddams, Th. P. Heavner, Th. E. Parker, S. R. Jefferts: The absolute frequency of the ^{87}Sr optical clock transition; *Metrologia* **45**, 539 (2008)

[13] X. Baillard, M. Fouché, R. Le Targat, P. G. Westergaard, A. Lecallier, F. Chapelet, M. Abgrall, G. D. Rovera, P. Laurent, P. Rosenbusch, S. Bize, G. Santarelli, A.

Clairon, P. Lemonde, G. Grosche, B. Lipphardt, H. Schnatz: An optical lattice clock with spin-polarized ^{87}Sr atoms; *Eur. Phys. J. D* **48**, 11 (2008)

[14] F.-L. Hong, M. Musha, M. Takamoto, H. Inaba, S. Yanagimachi, A. Takamizawa, K. Watabe, T. Ikegami, M. Imae, Y. Fujii, M. Amemiya, K. Nakagawa, K. Ueda, H. Katori: Measuring the frequency of a Sr optical lattice clock using a 120 km coherent optical transfer; *Opt. Lett.* **34**, 692 (2009)

[15] St. Falke, H. Schnatz, J. S. R. Vellore Winfred, Th. Middelmann, St. Vogt, S. Weyers, B. Lipphardt, G. Grosche, F. Riehle, U. Sterr and Ch. Lisdat: The ^{87}Sr optical frequency standard at PTB; *Metrologia* **48**, 399 (2011)

[16] A. Yamaguchi, N. Shiga, S. Nagano, Y. Li, H. Ishijima, H. Hachisu, M. Kumagai, T. Ido: Stability Transfer between Two Clock Lasers Operating at Different Wavelengths for Absolute Frequency Measurement of Clock Transition in ^{87}Sr; *Appl. Phys. Express* **5**, 022701 (2012)

[17] D. Akamatsu, H. Inaba, K. Hosaka, M. Yasuda, A. Onae, T. Suzuyama, M. Amemiya, F.-L. Hong: Spectroscopy and frequency measurement of the ^{87}Sr clock transition by laser linewidth transfer using an optical frequency comb; *Appl. Phys. Express* **7**, 012401 (2014)

[18] T. Tanabe, D. Akamatsu, T. Kobayashi, A. Takamizawa, S. Yanagimachi, T. Ikegami, T. Suzuyama, H. Inaba, S. Okubo, M. Yasuda, F.-L. Hong, A. Onae, K. Hosaka: Improved frequency measurement of the $^1S_0 - {}^3P_0$ clock transition in ^{87}Sr using a Cs fountain clock as a transfer oscillator; *J. Phys. Soc. Jpn.* **84**, 115002 (2015)

[19] Y.-G. Lin, Q. Wang, Y. Li, F. Meng, B.-K. Lin, E.-J. Zang, Z. Sun, F. Fang, T.-C. Li, Z.-J. Fang: First Evaluation and Frequency Measurement of the Strontium Optical Lattice Clock at NIM; *Chin. Phys. Lett.* **32**, 090601 (2015)

[20] S. Falke, N. Lemke, C. Grebing, B. Lipphardt, S. Weyers, V. Gerginov, N. Huntemann, C. Hagemann, A. Al-Masoudi, S. Häfner, S. Vogt, U. Sterr, C. Lisdat: A strontium lattice clock with 3 times $\times\ 10^{-17}$ inaccuracy and its frequency; *New J. Phys.* **16**, 073023 (2014)

[21] R. Le Targat, L. Lorini, Y. Le Coq, M. Zawada, J. Guéna, M. Abgrall, M. Gurov, P. Rosenbusch, D. G. Rovera, B. Nagórny, R. Gartman, P. G. Westergaard, M. E. Tobar, M. Lours, G. Santarelli, A. Clairon, S. Bize, P. Laurent, P. Lemonde, J. Lodewyck: Experimental realization of an optical second with strontium lattice clocks; *Nature Commun.* **4**, 2109 (2013)

[22] J. Lodewyck, S. Bilicki, E. Bookjans, J.-L. Robyr, C. Shi, G. Vallet, R. Le Targat, D. Nicolodi, Y. Le Coq, J. Guéna, M. Abgrall, P. Rosenbusch, S. Bize: Optical to microwave clock frequency ratios with a nearly continuous strontium optical lattice clock; *Metrologia* **53**, 1123 (2016)

[23] C. Grebing, A. Al-Masoudi, S. Dörscher, S. Häfner, V. Gerginov, S. Weyers, B. Lipphardt, F. Riehle, U. Sterr, C. Lisdat: Realization of a timescale with an accurate optical lattice clock; *Optica* **3**, 563 (2016)

[24] H. Hachisu, G. Petit, G., T. Ido: Absolute frequency measurement with uncertainty below 1×10^{-15} using International Atomic Time; *Appl. Phys. B* **123**, 34 (2017)

[25] H. Hachisu, G. Petit, F. Nakagawa, Y. Hanado, T. Ido: SI-traceable measurement of an optical frequency at the low 10^{-16} level without a local primary standard; *Optics Express* **25**, 8511 (2017)

[26] M. Chwalla, J. Benhelm, K. Kim, G. Kirchmair, T. Monz, M. Riebe, P. Schindler, A. S. Villar, W. Hänsel, C. F. Roos, R. Blatt, M. Abgrall, G. Santarelli, G. D. Rovera, Ph. Laurent: Absolute Frequency Measurement of the $^{40}Ca^{+}$ $4s\,^{2}S_{1/2} - 3d\,^{2}D_{5/2}$ Clock Transition; *Phys. Rev. Lett.* **102**, 023002 (2009)

[27] K. Matsubara, H. Hachisu, Y. Li, S. Nagano, C. Locke, A. Nogami, M. Kajita, K. Hayasaka, T. Ido, and M. Hosokawa: Direct comparison of a Ca^{+} single-ion clock against a Sr lattice clock to verify the absolute frequency measurement; *Optics Express* **20**, 22034 (2012)

[28] Y. Huang, H. Guan, P. Liu, W. Bian, L. Ma, K. Liang, T. Li, K. Gao: Frequency Comparison of Two $^{40}Ca^{+}$ Optical Clocks with an Uncertainty at the 10^{-17} Level; *Phys. Rev. Lett.* **116**, 013001 (2016)

[29] LNE-SYRTE TAI data; MJD 55954-57867 (Jan 2012 – April 2017); submitted on request of the CCL-CCTF WGFS by J. Guena on 10 May 2017

[30] Y. B. Ovchinnikov, K. Szymaniec and S. Edris: Measurement of rubidium ground-state hyperfine transition frequency using atomic fountains; *Metrologia* **52**, 595 (2015)

[31] J. Guéna, S. Weyers, M. Abgrall, C. Grebing, V. Gerginov, P. Rosenbusch, S. Bize, B. Lipphardt, H. Denker, N. Quintin, S. M. F. Raupach, D. Nicolodi, F. Stefani, N. Chiodo, S. Koke, A. Kuhl, F. Wiotte, F. Meynadier, E. Camisard, C. Chardonnet, Y. Le Coq, M. Lours, G. Santarelli, A. Amy-Klein, R. Le Targat, O. Lopez, P. E. Pottie, G. Grosche: First international comparison of fountain primary frequency standards via a long distance optical fiber link; *Metrologia* **54**, 348 (2017)

[32] K. Yamanaka, N. Ohmae, I. Ushijima, M. Takamoto and H. Katori: Frequency Ratio of ^{199}Hg and ^{87}Sr Optical Lattice Clocks beyond the SI Limit; *Phys. Rev. Lett.* **114**, 230801 (2015)

[33] D. Akamatsu, M. Yasuda, H. Inaba, K. Hosaka, T. Tanabe, A. Onae, F.-L. Hong: Frequency ratio measurement of ^{171}Yb and ^{87}Sr optical lattice clocks; *Optics Express* **22**, 7898 (2014)

[34] D. Akamatsu, M. Yasuda, H. Inaba, K. Hosaka, T. Tanabe, A. Onae, F.-L. Hong: Errata: Frequency ratio measurement of ^{171}Yb and ^{87}Sr optical lattice clocks; *Optics Express* **22**, 32199 (2014)

[35] M. Takamoto, I. Ushijima, M. Das, N. Nemitz, T. Ohkubo, K. Yamanaka, N. Ohmae, T. Takano, T. Akatsuka, A. Yamaguchi, H. Katori: Frequency ratios of Sr, Yb, and Hg based optical lattice clocks and their applications; *C. R. Physique* **16**, 489 (2015)

[36] N. Nemitz, T. Ohkubo, M. Takamoto, I. Ushijima, M. Das, N. Ohmae, H. Katori: Frequency ratio of Yb and Sr clocks with 5×10^{-17} uncertainty at 150 seconds averaging time; *Nature Photon.* **10**, 258 (2016)

[37] T. Takano, R. Mizushima, H. Katori: Precise determination of the isotope shift of ^{88}Sr—^{87}Sr optical lattice clock by sharing perturbations; *Appl. Phys. Express* **10**, 072801 (2017)

[38] T. Kohno, M. Yasuda, K. Hosaka, H. Inaba, Y. Nakajima, F. L. Hong: One-Dimensional Optical Lattice Clock with a Fermionic ^{171}Yb Isotope; *Appl. Phys. Express* **2**, 072501 (2009)

[39] M. Yasuda, H. Inaba, T. Kohno, T. Tanabe, Y. Nakajima, K. Hosaka, D. Akamatsu, A. Onae, T. Suzuyama, M. Amemiya, F.-L. Hong: Improved Absolute Frequency Measurement of the ^{171}Yb Optical Lattice Clock towards a Candidate for the Redefinition of the Second; *Appl. Phys. Express* **5**, 102401 (2012).

[40] N. D. Lemke, A. D. Ludlow, Z.W. Barber, T. M. Fortier, S.A. Diddams, Y. Jiang, S. R. Jefferts, T. P. Heavner, T. E. Parker, and C.W. Oates: Spin-½ Optical Lattice Clock; *Phys. Rev. Lett.* **103**, 063001 (2009)

[41] C. Y. Park, D. H. Yu, W.-K. Lee, S. E. Park, E. B. Kim, S. K. Lee, J. W. Cho, T. H. Yoon, J. Mun, S. J. Park, T. Y. Kwon and S.-B. Lee: Absolute frequency measurement of $^1S_0(F=1/2) - {}^3P_0(F=1/2)$ transition of ^{171}Yb atoms in a one-dimensional optical lattice at KRISS; *Metrologia* **50**, 119-128 (2013)

[42] M. Pizzocaro, P. Thoumany, B. Rauf, F. Bregolin, G. Milani, C. Clivati, G. A. Costanzo, F. Levi, D. Calonico: Absolute frequency measurement of the $^1S_0 - {}^3P_0$ transition of ^{171}Yb; *Metrologia* **54**, 102 – 112 (2017)

[43] H. Kim, M.-S. Heo, W.-K. Lee, C.Y. Park, H.-G. Hong, S.-W. Hwang and D.-H. Yu: Improved absolute frequency measurement of the ^{171}Yb optical lattice clock at KRISS relative to the SI second; *Jpn. J. Appl. Phys.* **56** 050302 (2017)